

Package #164

Replace the active heparin coating with a passive, ultra-smooth, low-surface-energy polymer lining (e.g., fluorinated polymer) that physically resists protein and platelet adhesion, thereby limiting clot initiation without releasing any drug.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter flow occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency through occlusion despite flushing?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 100000: Injury [VITAL SIGNS]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Contact activation of coagulation in newly inserted central venous catheters (CAS-2): protocol for a randomised controlled study comparing four commercially available catheters.'; 'Nursing management of a patient with calf muscle venous thrombosis complicated by bladder clot retention following oocyte retrieval: A case report.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Anticoagulant heparin coating on the catheter surface reduces thrombus formation, leading to longer patency.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
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Status: ENGINEERING_DEFINITION. Not physically validated.